1. Introduction

Green Cart Ltd. is a UK-based e-commerce company specializing in eco-friendly household products. The objective of this project was to analyze sales and customer behavior data to support the Q2 performance review. The analysis focused on revenue trends, product performance, customer loyalty value, discount effectiveness, delivery performance, and payment behavior.
Three datasets were used: sales data, product information, and customer information. The goal was to clean, merge, and analyze the data to generate insights that inform marketing, operational, and strategic decisions.

2. Data Cleaning Summary

Before analysis, the datasets required structured cleaning to ensure accuracy.

For the **sales dataset**, the quantity column was converted to numeric format and order_date was converted to datetime. Text fields such as delivery_status, payment_method, delayed and region were standardized using trimming and consistent casing. Missing values in discount_applied were filled with 0, assuming no discount was provided. Records missing critical identifiers (customer_id or product_id) were removed to preserve merge integrity. Duplicate orders were removed using order_id, and numeric fields were validated to ensure non-negative values.

For the **product dataset**, launch_date was converted to datetime, and text fields were standardized.

For the **customer dataset**, signup_date was converted to datetime, and categorical fields such as gender, region, and loyalty_tier were cleaned and standardized. Missing categorical values were replaced with “Unknown”.

After merging, additional inconsistencies in loyalty tiers, regions, and payment methods were corrected to ensure accurate aggregation.

These steps ensured consistent analysis and prevented inaccurate results in later stages.
It also ensured reliable aggregation and accurate trend analysis.

3. Feature Engineering

Several new features were created to enhance business insights:

- **Revenue:** Calculated as quantity × unit price × (1 – discount applied) to measure actual transaction value.

[2]:

	quantity	unit_price	discount_applied	revenue
0	3.0	39.25	0.00	117.750
1	5.0	18.92	0.00	94.600
2	1.0	29.68	0.15	25.228
3	1.0	32.76	0.20	26.208
4	1.0	47.62	0.20	38.096

- **Order Week:** Extracted ISO week to analyze weekly performance trends.

	order_date	order_week
0	2025-06-07	23
1	2025-06-07	23
2	2025-07-07	28
3	2025-08-07	32
4	2025-08-07	32

- **Price Band:** Categorized products into Low, Medium, and High price segments.

	unit_price	price_band
0	39.25	High
1	18.92	Medium
2	29.68	Medium
3	32.76	High
4	47.62	High

- **Days to Order:** Measured time between product launch and purchase to evaluate lifecycle performance.

	launch_date	order_date	days_to_order
0	2024-04-10	2025-06-07	423.0
1	2025-01-18	2025-06-07	140.0
2	2025-03-25	2025-07-07	104.0
3	2024-07-15	2025-08-07	388.0
4	2025-02-20	2025-08-07	168.0

- **Email Domain:** Extracted from email addresses for behavioral segmentation.

	email	email_domain
0	jacquelinemartin@mills-logan.com	mills-logan.com
1	georgeweber@morgan.com	morgan.com
2	lcook@walters-smith.com	walters-smith.com
3	smoore@gmail.com	gmail.com
4	mcclainkatherine@hotmail.com	hotmail.com

```
email_domain
yahoo.com       576
hotmail.com     466
gmail.com       462
Unknown          55
young.com        17
thomas.com       17
smith.com        16
goodman.info     15
jones.com        13
davis.com        13
Name: count, dtype: int64
```

- **Is Late:** Binary flag identifying delayed deliveries.

	delivery_status	is_late
0	Delivered	False
1	Delayed	True
2	Delivered	False
3	Cancelled	False
4	Delayed	True

- **Signup Month:** Extracted month of customer registration to measure purchase patterns.

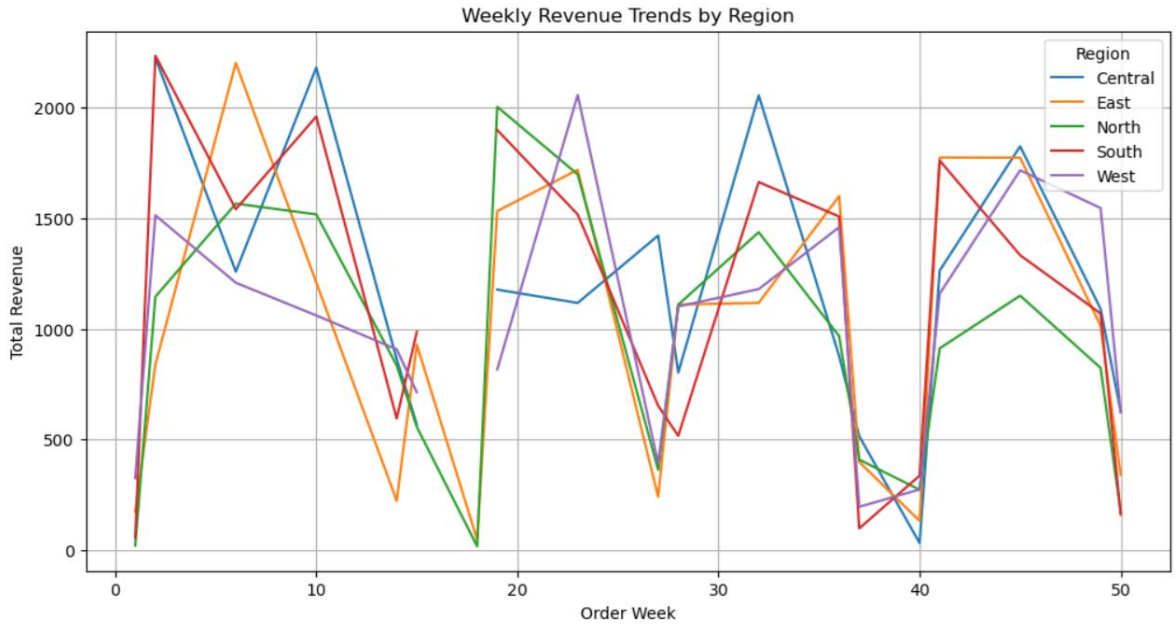
loyalty_tier	Bronze	Gold	Silver	Unknown
signup_month				
Apr	28	145	69	0
Aug	49	147	47	0
Dec	59	150	23	0
Feb	61	111	53	6
Jan	40	136	52	0
Jul	47	129	34	0
Jun	81	101	64	0
Mar	57	95	83	0
May	41	112	54	3
Nov	46	153	62	0
Oct	73	172	75	0
Sep	33	201	41	0

These engineered features allowed deeper trend, operational, and behavioral analysis.

4. Key Findings & Trends

Revenue Performance

Revenue fluctuates week-to-week, with Week 23 generating the highest total revenue (£8,105.78). The Central region establishes consistently strong performance, although revenue direction varies by week. The pattern suggests campaign-driven or seasonal demand rather than steady linear growth.

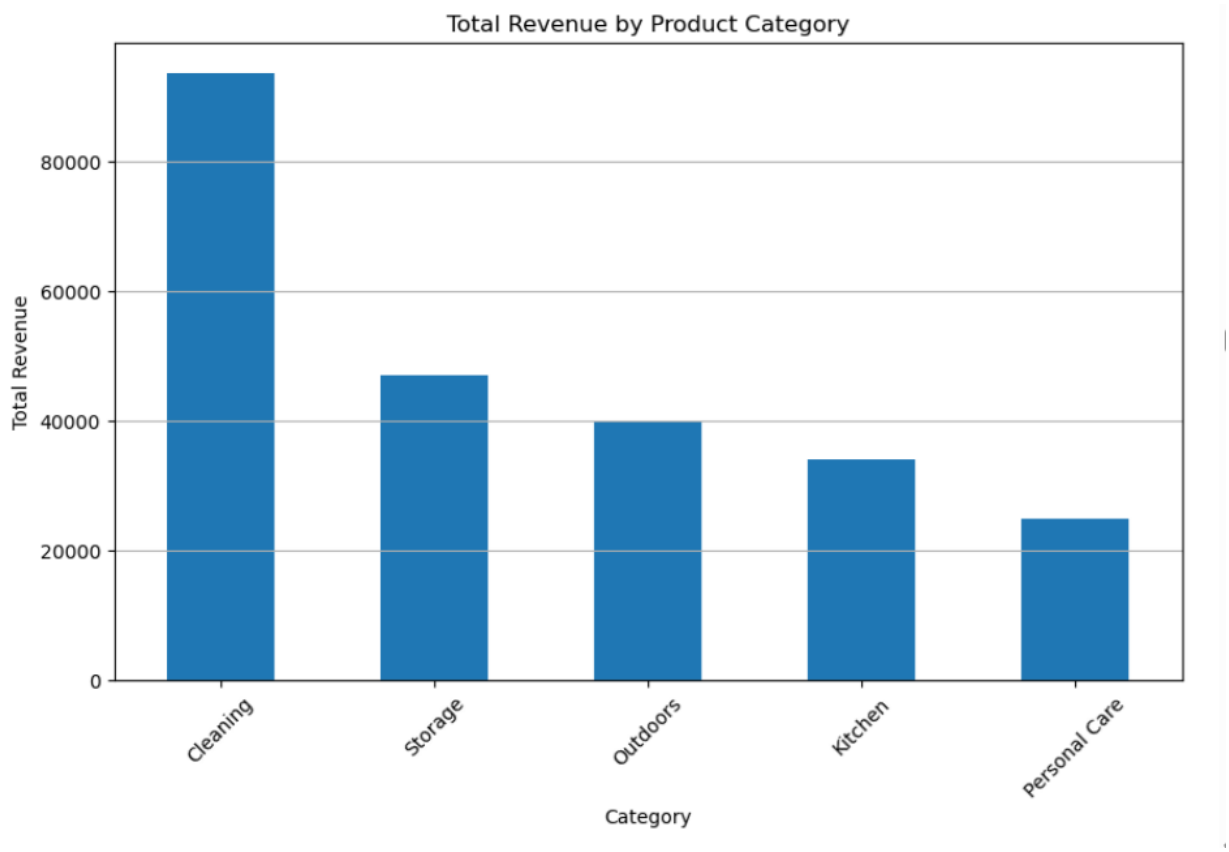


Product Category Performance

The Cleaning category significantly beats others, generating over £93,000 in revenue and the highest sales volume. This indicates strong inherent demand rather than dependance on heavy discounts.

Storage and outdoors form the mid-tier performers, while Personal Care contributes comparatively less revenue despite moderate discount levels.

category	total_revenue	total_quantity	avg_discount
Cleaning	93567.9475	3581.0	0.085595
Storage	46931.4575	1730.0	0.081043
Outdoors	40062.0680	1519.0	0.082016
Kitchen	33933.6760	1226.0	0.075622
Personal Care	24892.2765	900.0	0.087043



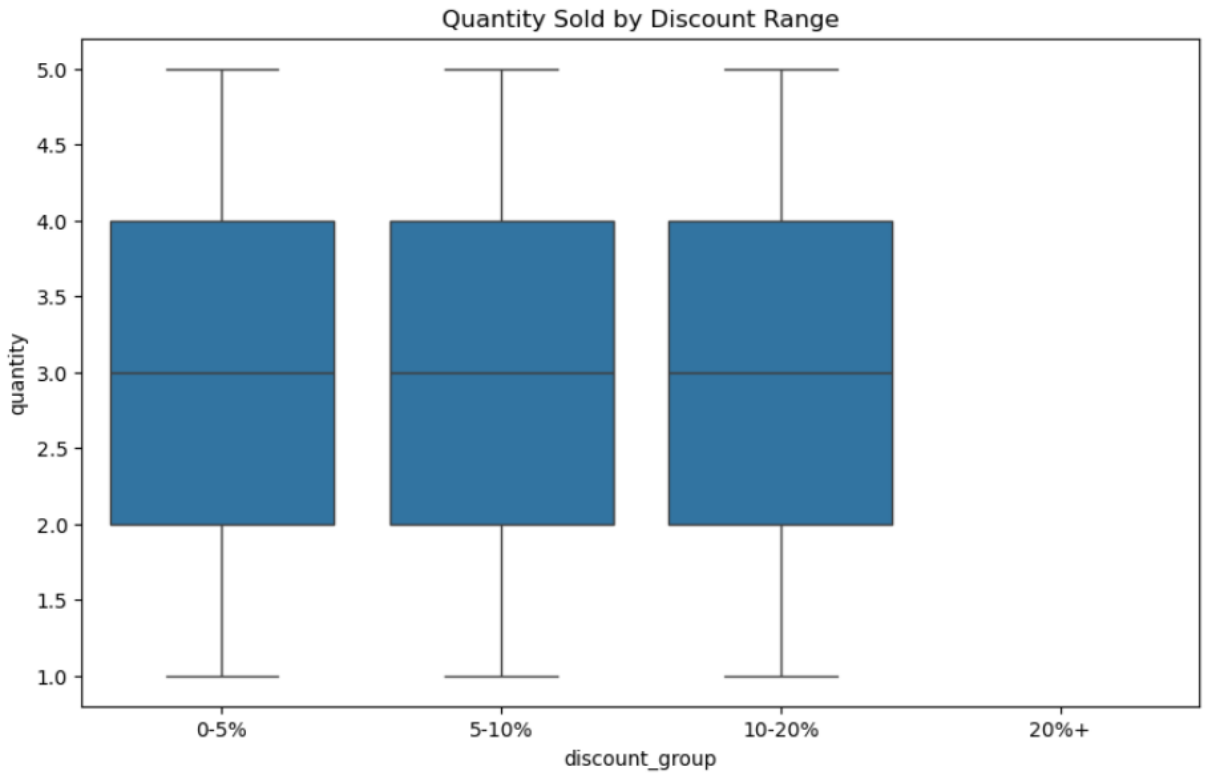
Discount Effectiveness

Statistical analysis shows no meaningful correlation between discount levels and quantity sold (correlation ≈ -0.01). Orders with minimal discounts (0–5%) generate the highest average revenue per transaction.

Higher discount ranges reduce average revenue without increasing sales. This indicates that discounts do not significantly lead demand and may reduce profitability.

```
discount_group
0-5%      87.032439
5-10%     77.655202
10-20%    71.802598
20%+            NaN
Name: revenue, dtype: float64
```

```
discount_group
0-5%      3.085828
5-10%     2.982213
10-20%    2.968560
20%+           NaN
Name: quantity, dtype: float64
```



Loyalty Tier Value

Gold loyalty members generate the highest value:

- £136,177 total revenue
- 1,670 orders
- £81.54 average revenue per order

The difference in average order value between tiers is small; however, Gold customers purchase more frequently. Gold customers dominate all regions, demonstrating consistent nationwide engagement.

Silver and Bronze tiers contribute steadily but remain significantly below Gold in total impact.

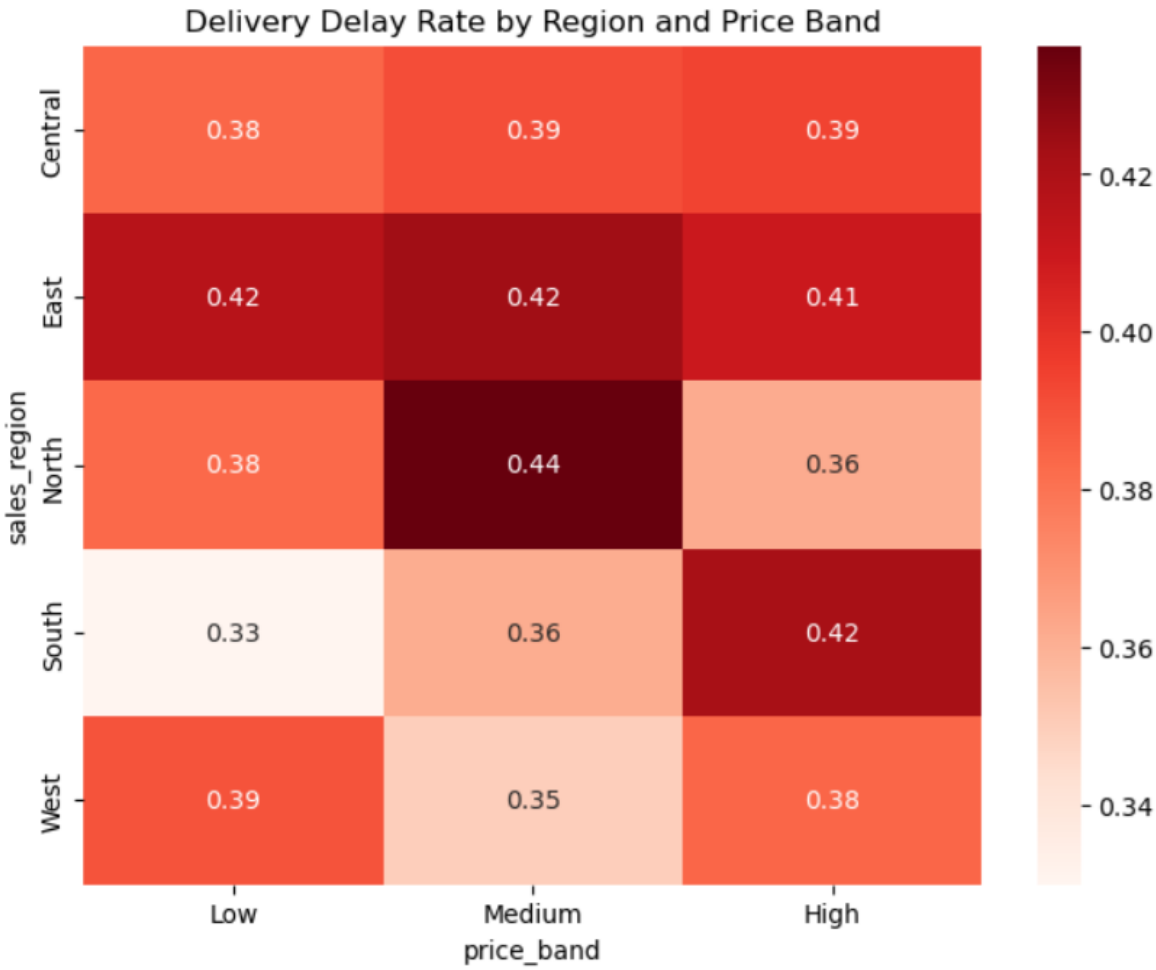
loyalty_tier	total_revenue	total_quantity	avg_revenue	order_count
Gold	136177.1300	5020.0	81.543192	1670
Silver	52032.6080	1973.0	79.197272	657
Bronze	49085.0685	1864.0	78.536110	624
Unknown	2092.6190	99.0	63.412697	33

Delivery Performance

Delivery performance presents a clear operational concern.
Overall delay rate is approximately 39%. The highest delay rates were observed in:

- North – Medium price band (44%)
- East – Medium price band (42%)
- South – High price band (42%)

The East region shows consistently high delays across all price bands which suggests basic fulfilment inefficiencies. High-value products also show elevated delay risk, possibly due to inventory or logistics limits.

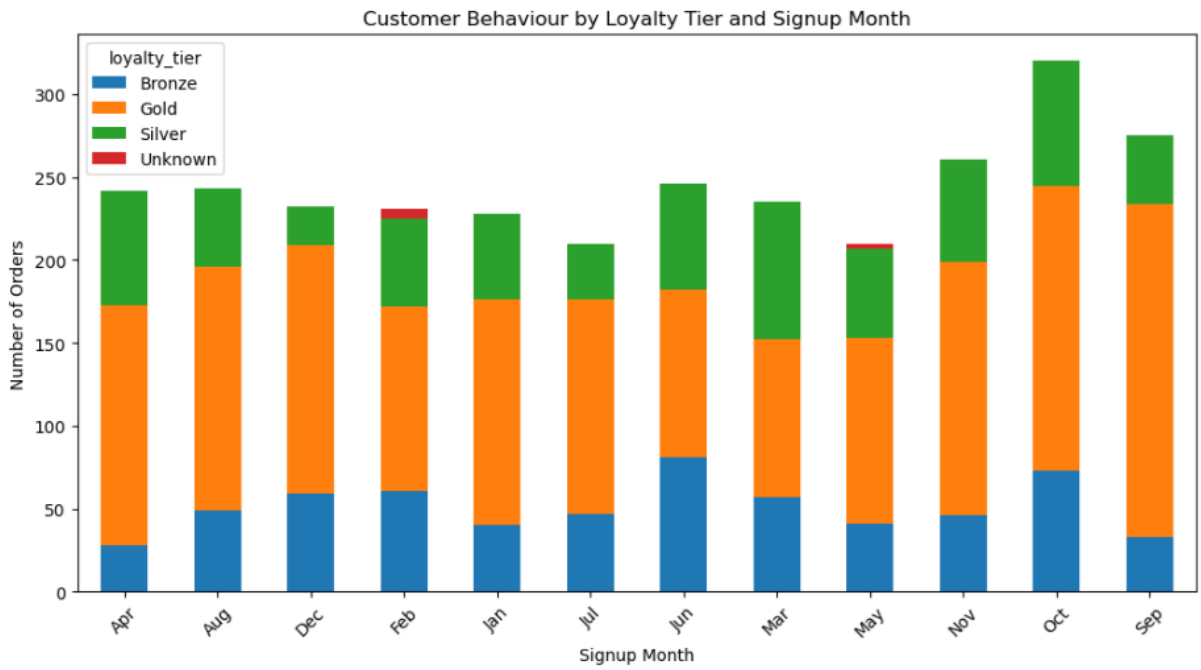

Delivery Delay Rate by Region and Price Band
sales_region
Central
East
North
South
West
0.38
0.39
0.39
0.42
0.42
0.41
0.38
0.44
0.36
0.33
0.36
0.42
0.39
0.35
0.38
Low
Medium
High
price_band
0.42
0.40
0.38
0.36
0.34

sales_region	price_band	total_orders	delayed_orders	delay_rate
North	Medium	204	89	0.436275
East	Medium	236	100	0.423729
South	High	294	124	0.421769
East	Low	96	40	0.416667
	High	266	109	0.409774
Central	High	264	104	0.393939
	Medium	225	88	0.391111
West	Low	77	30	0.389610
	High	268	103	0.384328
Central	Low	112	43	0.383929
North	Low	107	41	0.383178
	High	293	106	0.361775
South	Medium	202	73	0.361386
West	Medium	243	85	0.349794
South	Low	97	32	0.329897

Customer Signup Behavior

Gold customers dominate purchasing activity across all signup months. Q4 (September–December) shows stronger activity levels, indicating seasonal demand patterns.

Customer purchase appears consistently across months, with no extreme instability. The loyalty program demonstrates stable geographic and temporal performance.

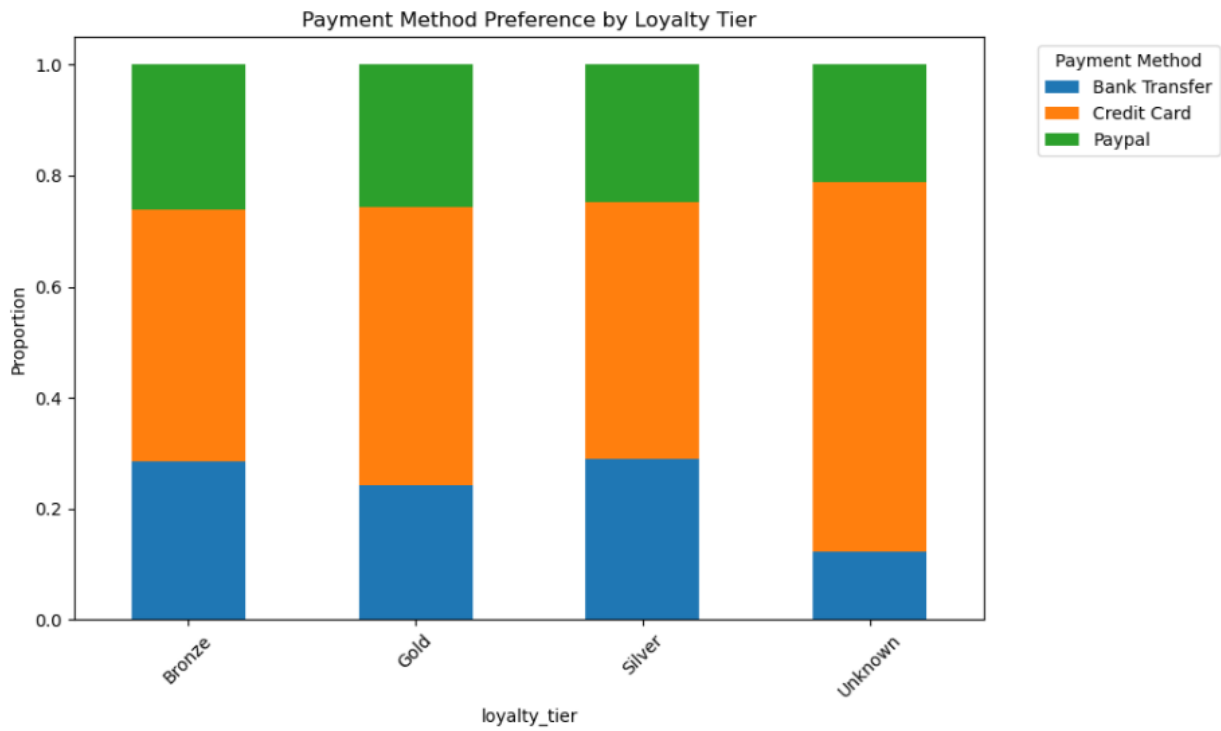


Payment Behavior

Credit Card is the major payment method across all loyalty tiers. Payment preferences remain moderately consistent across bronze, silver, and gold customers. There is no significant behavioral division based on payment method.

This suggests strong digital maturity and low dependency on alternative payment channels.

payment_method	Bank Transfer	Credit Card	Paypal
loyalty_tier			
Bronze	0.284800	0.454400	0.260800
Gold	0.241607	0.502998	0.255396
Silver	0.289634	0.463415	0.246951
Unknown	0.121212	0.666667	0.212121



5. Business Question Responses

Which product categories drive the most revenue?

- The ‘Cleaning category’ drives the highest revenue and volume, particularly across multiple regions.

Do discounts lead to more items sold?

- No. There is no meaningful relationship between discount levels and sales volume.

Which loyalty tier generates the most value?

- Gold customers generate the most revenue, primarily due to higher purchase frequency.

Are certain regions struggling with delivery delays?

- Yes. North and East show elevated delay rates, particularly in Medium and High price bands.

Do customer signup patterns influence purchasing activity?

- Seasonal patterns suggest stronger purchasing in Q4, but loyalty dominance remains stable across months.

Recommendations

1. **Reduce aggressive discounting strategies.**
   Instead than reducing margins through price cuts, focus on value-based marketing.

2. **Investigate logistics performance in the East and North regions.**
   High delay rates point to potential inefficiencies in achieving goals.

3. **Improve Gold loyalty management and tier upgrade paths.**
   Gold customers drive sustainable revenue growth.

4. **Prepare inventory and marketing initiatives for Q4 peaks.**
   Periodic demand trends should guide operational planning.

7. Data Risks and Improvements

Key data quality issues included inconsistent categorical labels and lifecycle inconsistencies. Implementing standardized input validation, dropdown-based category controls, and automated data checks at source would significantly reduce manual correction needs.

Conclusion

The analysis demonstrates that Green Cart Ltd. is strongly supported by loyal Gold customers and high-performing cleaning products. However, discount strategies do not immensely increase sales. Delivery delays present operational risk. Addressing logistics inefficiencies and improving loyalty strategy will increase both profitability and customer satisfaction.